

Wellspring Clinics

Power BI Data Model, Dashboard & Analysis

Project Type:	Power BI Data Model & Dashboard
Tool Used:	Microsoft Power BI Desktop
Dataset:	Excel workbook — Clinics, Patients, Staff, Visits (4 related tables)

1. Project Overview

This project involved building a Power BI data model and dashboard for Wellspring Clinics, a multi-site healthcare provider. The underlying dataset consisted of four related tables — Clinics, Patients, Staff, and Visits — supplied as an Excel workbook. The work covered structuring the data model, adding a derived fee classification, resolving data quality issues, and building a dashboard to summarize visit, fee, and staff experience patterns.

2. Dataset Structure

Table	Key Field(s)	Purpose
Clinics	ClinicID	One record per clinic: name, state, type, date opened
Patients	PatientID	One record per patient: demographics and registration date
Staff	StaffID, ClinicID	One record per staff member, linked to the clinic they work at
Visits	VisitID, PatientID, ClinicID, StaffID	One record per patient visit — the fact table connecting patients, clinics, and staff

Visits is the central "fact" table: it is the only table where PatientID and ClinicID appear together, and it links out to Patients, Clinics, and Staff through foreign keys.

3. Fee Classification

A calculated column, **FeeCategory**, was added to the Visits table to classify each visit's consultation fee as "Expensive" ($\geq 10,000$) or "Affordable" ($< 10,000$).

Data Quality Note: ConsultationFee was originally stored as text, in mixed formats (e.g. "■3,000" and "NGN 3000"). Before it could be classified correctly, the column was cleaned in Power Query — currency symbols and thousands separators removed, and the field converted to a numeric data type.

4. Data Modeling — Avoiding Silent Errors

An unconnected table in a Power BI model — one with no relationship linking it to the rest of the model — is a common and easy-to-miss mistake:

- Slicers and filters from other tables have no effect on visuals built from an unconnected table, and vice versa.
- Measures meant to aggregate "by clinic" or "by patient" instead return unfiltered totals, since there's no relationship for the filter to travel across.
- The error is silent — the report still runs and displays numbers, so the mistake can go unnoticed until someone checks whether the figures actually respond to filtering.

In this model, the risk case was **Staff**, which only connects to the rest of the model through ClinicID. Confirming that relationship stayed active was essential — if it were missing, filtering the report by clinic would leave staff-based visuals unchanged.

5. Choosing the Right Table for "Unique Patients per Clinic"

Recommendation: build the visual from Visits, not Clinics or Patients.

Table	Why it does / doesn't work
Clinics	Has no patient information at all — nothing to count.
Patients	Has no clinic information — a patient isn't tied to any clinic until a visit occurs.
Visits	The only table where PatientID and ClinicID occur together, making the relationship meaningful.

A distinct count (rather than a plain count) was used, since a patient can visit the same clinic more than once — a plain count would overstate the number of unique patients.

6. Data Cleaning Decisions

Of the 422 total patients who visited the clinics, were diagnosed, and paid consultation fees, 18 patients had no wait time recorded. All other fields were complete for these 18 records, suggesting the wait time was either not logged at the point of care or lost during data entry.

Decision: these 18 records were flagged with a "Not Recorded" category rather than removed, preserving the rest of their data for analysis.

The field originally named "Follow Up Required" was renamed to "Follow Up Required Medication" for clarity on the dashboard.

7. Dashboard Overview

The Power BI dashboard brings together clinic operations, fees, and staffing into one view, with slicers for Department and Staff Name so any visual can be filtered interactively.

Visual	Metric / Content
Average Wait Time (card)	35.25 minutes
Total Consultation Fee (card)	2,853,000

Visual	Metric / Content
Total Diagnostics (gauge)	422, within a range of 210–844
Follow Up Required Medications	Yes / No filter checkboxes
Wait Time in Minutes (area chart)	Distribution of wait times from 0–100 minutes, with most visits clustering under 50 minutes
Total Consultation Fee by Department (line chart)	Declining trend from Immunization (highest) to Antenatal (lowest)
Years of Experience of Each Staff (funnel)	24 staff members ranked from 25 years (Aisha Suleiman) down to 1 year (Hauwa Abubakar)

8. Key Findings

- Immunization had the highest sum of consultation fees at 470,500 — 32.16% higher than Antenatal, the lowest at 356,000.
- Immunization accounted for 16.49% of the total consultation fees across all departments.
- Across all 7 departments, total consultation fees ranged from 356,000 to 470,500 — a relatively narrow spread.
- Total diagnostics across the period was 422, with average wait time sitting at 35.25 minutes.
- Staff experience varies widely — from 25 years (Aisha Suleiman, the most experienced) down to 1 year (Hauwa Abubakar, the newest), with most staff clustered between 3 and 13 years.

9. Summary

- Visits is the analytical core of this model — most cross-table questions (fees, patient counts, staff activity) resolve through it.
- Data types and formats (especially ConsultationFee) needed cleaning before fee classification and department-fee comparisons would be accurate.
- Missing wait-time values were handled by flagging rather than deleting, keeping the dataset complete for other analyses.
- Immunization drives the largest share of consultation fees, while Antenatal generates the least — a pattern worth investigating further (e.g. visit volume vs. per-visit fee).
- Every table in a data model should be checked for an active relationship — an isolated table is a silent source of incorrect filtering.